PCB Motor Rev1

Assembly Master Bill of Materials and Procurement Checklist

Configuration	S1 - R1 - S2 - R2
Stators	Two stator stacks, five 1.0 mm PCBs per stack
Rotors	R1 double-sided shared rotor; R2 inward-facing single-sided rotor
Magnets	48 installed from 60 available; 12 retained as spares
Rotor back iron	2 mm low-carbon mild steel - thickness fixed
Document purpose	Workshop fabrication, purchasing, assembly preparation and laboratory testing

Prepared for the EEE4022S modular axial-flux PCB motor prototype. Dimensions identified as provisional must be confirmed by electromagnetic optimisation and the final axial-stack CAD before manufacture.

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1. Design status and purchasing rules

This document separates fixed specifications from provisional dimensions. A provisional value is suitable for modelling, quotations or layout studies, but should not be treated as a final machining dimension.

Parameter	Current value	Status / action
Rotor steel thickness	2 mm	FIXED
Magnets	N35, 20 x 6 x 2 mm	FIXED - already received
PCB thickness	1.0 mm	FIXED - 10 boards ordered
PCBs per stator	5	FIXED
Shaft diameter	8 mm	CURRENT DESIGN - confirm after load calculation
Bearing envelope	608-2RS, 8 x 22 x 7 mm	RECOMMENDED - verify axial-load capacity
Rotor outside diameter	86 mm baseline	PROVISIONAL - optimise/check mechanical margin
Magnet centre radius	28 mm baseline	PROVISIONAL - optimise approximately 27-29 mm
Mechanical air gap	1.0 mm baseline	ADJUSTABLE - study approximately 0.5-1.5 mm
Rotor pole pattern	16 alternating poles	FIXED
Installed Rev1 magnets	48	FIXED for S1-R1-S2-R2

MAGNET COUNT For a sequential modular build, 60 magnets are sufficient: Rev1 uses 48 and leaves 12 spares. If Rev0 and Rev1 must remain assembled at the same time, 64 magnets are required before spares, so at least 4 additional magnets would be needed.

2. Main electromagnetic and rotor components

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Stator PCBs**	10	Six-layer FR-4; 1.0 mm thick; approximately 93 mm across flats; 25 mm centre opening; 12 x 3.1 mm mounting holes on 90 mm PCD; 1 oz copper; OSP finish.	Five PCBs form each stator stack. Copper tracks act as the three-phase windings.	Already ordered. Replacement fabrication: JLCPCB
[ ]	**Neodymium magnets**	48 used	N35; 20 x 6 x 2 mm; magnetised through the 2 mm thickness; Ni-Cu-Ni coating.	Provide the rotor field. Arrange 16 alternating N/S poles on each active rotor face.	Already received. Reference: UMAGNETS
[ ]	**Spare magnets**	12	Same batch, grade, size and magnetisation direction as installed magnets.	Allow for damaged plating, polarity mistakes, adhesive trials and future replacement.	Already received; store with nonmagnetic separators.
[ ]	**Rotor back-iron discs**	2	Low-carbon mild steel; 2 mm thick; baseline 86 mm OD. Centre pilot and mounting pattern to match final hub CAD.	Carry magnets, provide magnetic return path and transmit torque into each hub.	Fabricate through Power Lab/UCT workshop or local laser/waterjet service.
[ ]	**Magnet-retention plates**	3	PETG, nylon or glass-filled nylon; approximately 2.1-2.5 mm thick; 16 pockets. Initial pocket size about 20.2 x 6.2 mm; radial location remains provisional.	Mechanically capture magnets so centrifugal retention does not depend only on adhesive.	3D print at UCT or local additive-manufacturing service.
[ ]	**Split-clamp rotor hubs**	2	8 mm bore; flanged interface to steel rotor. Final pilot diameter, flange diameter, bolt size and PCD from completed CAD.	Hold each thin rotor concentrically on the shaft without marking it with a grub screw.	Custom machine at UCT; adapt a commercial clamp hub only after dimension check.
[ ]	**Rotor spacers**	As req.	Precision aluminium, stainless steel or POM rings; approximately 8.1 mm bore; thicknesses from final axial drawing.	Set axial positions of rotors, bearings and collars.	Machine locally or use precision shim rings.

Magnet allocation

Rotor face	Magnets	Arrangement
R1 face toward S1	16	Alternating N/S around circumference
R1 face toward S2	16	Shared rotor second active face
R2 face toward S2	16	Only the inward-facing side is populated
Total installed	48	Leaves 12 spares from the 60 purchased

3. Shaft, bearings and rotor support

The shaft length, bearing spacing and coupling position must be taken from one complete axial cross-section before the shaft is cut.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Precision shaft**	1	8 mm diameter, preferably h6 tolerance; precision-ground steel; buy approximately 300 mm and cut after axial stack is fixed.	Supports both rotors, passes through bearings and transfers power to the load.	BMG, RS or local engineering/bearing supplier.
[ ]	**Spare shaft**	1 rec.	Same 8 mm material and stock length as main shaft.	Provides recovery if the first shaft is bent, scored or cut too short.	Same source as main shaft.
[ ]	**Sealed bearings**	2 + 2 spare	608-2RS; 8 mm bore x 22 mm OD x 7 mm wide. Confirm the 2RS rubber-seal suffix.	Provide radial shaft support and environmental protection. Verify axial magnetic force before relying on them for continuous thrust load.	BMG or local bearing supplier.
[ ]	**Bearing housings**	2	Custom aluminium blocks with nominal 22 mm seats; workshop to specify final tolerance/fit and retention method.	Keep the shaft axis aligned and locate the bearings on the base.	Machine at Power Lab/UCT workshop.
[ ]	**Clamp shaft collars**	4	8 mm bore; one-piece or two-piece clamp type; typical width 8-9 mm. Avoid point-contact grub-screw collars where possible.	Set axial position and prevent the shaft/rotor assembly moving through the bearings.	RS shaft collars
[ ]	**Flexible coupling**	1	Motor side 8 mm bore; load side selected to match torque sensor, dynamometer or load shaft.	Transmits torque while tolerating small angular and lateral misalignment.	RS, robotics supplier or local coupling supplier.
[ ]	**Axial/thrust support**	If req.	Thrust bearing, angular-contact bearing pair or designed shaft shoulder; select after force simulation.	Carries substantial continuous axial magnetic force if the 608 bearings are inadequate.	Final selection after Maxwell axial-force result.

4. Stator-stack clamping hardware

Preliminary bolt length: two clamp-plate thicknesses + five PCB thicknesses + washers + full nut engagement. Order final lengths only after measuring a real stack.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Stator clamp plates**	4	Two per stator; 2-3 mm aluminium, PETG, nylon or G10. Clear all tracks, joints and terminals; centre opening at least 25 mm.	Compress each five-PCB stack into one stable stator without obstructing the active winding region.	Laser cut, CNC machine or 3D print.
[ ]	**Compression sleeves**	12	Six per stator; about 3.2 mm ID. Exact length determined by actual PCB stack and clamp design.	Carry bolt compression and prevent bowing/crushing of PCBs.	Machine from POM, nylon or aluminium tube.
[ ]	**M3 stator bolts**	12 + 6 spare	M3; likely 20-25 mm, but calculate after measuring one complete stack.	Pass through six selected mounting holes to clamp each stator assembly.	Boltfast, RS, Builders or local fastener supplier.
[ ]	**M3 nyloc nuts**	18	M3, matching bolt material.	Resist loosening under vibration while remaining removable.	Local fastener supplier.
[ ]	**M3 nylon washers**	36-48	Small outside diameter; check clearance to copper/features.	Protect and electrically isolate the PCB faces around mounting holes.	RS or electronics supplier.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**PCB-safe spacers**	As req.	Nylon/POM; height determined from solder-joint and terminal clearances.	Prevent one PCB from pressing on raised features of another.	Electronics supplier or 3D print.
[ ]	**Stator locating pins**	4-6 rec.	About 3 mm diameter; electrically safe/nonmagnetic material preferred.	Improve angular registration of the five PCBs in each stack.	Machine locally.

5. Frame and air-gap adjustment

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Rigid baseplate**	1	Approximately 6-10 mm aluminium plate or rigid extrusion frame. Final plan dimensions from complete assembly layout.	Provides a common datum for bearings, stators, coupling and load.	UCT workshop or aluminium supplier.
[ ]	**Bearing brackets**	2	Rigid aluminium/steel with slotted base mounting recommended.	Carry bearing housings and permit shaft alignment.	Custom fabricate.
[ ]	**Adjustable stator frames**	2	Nonmagnetic material preferred close to active field; slotted axial mounting.	Hold S1 and S2 independently while permitting controlled air-gap adjustment.	Custom fabricate.
[ ]	**Axial adjustment screws**	4-8	M4 or M5, fine pitch preferred, with locknuts.	Provide controlled stator movement during gap setting.	Fastener supplier.
[ ]	**Precision shim kit**	1	Include 0.1, 0.2, 0.5, 0.75, 1.0 and 1.5 mm thicknesses where available.	Creates repeatable stator-to-rotor spacings for tests.	RS or engineering supplier.
[ ]	**Metric feeler gauges**	1	Cover at least 0.05-1.0 mm.	Measure the air gap at several angular positions and reveal rotor runout/tilt.	Local tool supplier.
[ ]	**Levelling feet**	4	M6 or M8 adjustable feet, preferably rubber ended.	Level and stabilise the test rig.	Hardware or automation supplier.
[ ]	**Rubber isolation pads**	4	Approximately 3-6 mm thick.	Reduce vibration transferred into the bench.	Hardware supplier.
[ ]	**Mechanical end stops**	2	Extra 8 mm collars or dedicated stops outside bearings.	Prevent the shaft leaving the structure if an axial collar loosens.	Use spare clamp collars or custom stops.

6. Rotor fasteners and retention hardware

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Hub-to-rotor bolts**	8-12	M3, probably 8-12 mm long; final length from flange/rotor/retainer stack.	Fasten each steel rotor disc to its hub.	Local fastener supplier.
[ ]	**Hub clamp bolts**	2-4	M3 or M4 as required by the split-hub design.	Close the hub clamp uniformly around the shaft.	Local fastener supplier.
[ ]	**Retention-plate screws**	12-24	M2.5 or M3; nonmagnetic stainless preferred where practical.	Mechanically attach the magnet-retention plates.	RS or local fastener supplier.
[ ]	**Nuts/threaded inserts**	1 per screw	Nyloc nuts, heat-set inserts or tapped hub interface, selected with the CAD.	Provide secure, serviceable retention-plate threads.	Fastener/electronics supplier.
[ ]	**Medium threadlocker**	1 bottle	Loctite 243 or equivalent removable medium-strength grade.	Stops metal fasteners loosening under vibration while preserving serviceability.	Industrial/hardware supplier.
[ ]	**M3/M4 washers**	Assorted	Flat washers matched to all bolts.	Spread clamp load and protect component surfaces.	Fastener supplier.
[ ]	**8 mm shim washers**	Assorted	Several controlled thicknesses.	Fine adjustment of rotor and collar axial positions.	Engineering supplier.

7. Magnet installation and consumables

Do not depend on a fast-setting repair epoxy for the entire 16-magnet operation. The selected adhesive must pass a coupon test using the actual steel preparation and magnet coating.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Structural epoxy**	2 packs	Slow-curing, toughened two-part epoxy compatible with nickel-plated magnets and mild steel; validate using coupons.	Bonds magnets to steel while retention plates provide mechanical capture. Slow cure gives working time for 16 positions.	Pratley or industrial adhesive supplier.
[ ]	**Adhesive test coupons**	6-10	Actual 2 mm steel offcuts plus representative nickel-plated surfaces.	Verify cleaning, surface preparation, bond strength and cure before rotor assembly.	Retain rotor offcuts from fabrication.
[ ]	**Isopropyl alcohol**	500 mL	Preferably 99% IPA.	Degreases prepared steel and magnet surfaces.	Pharmacy, electronics or chemical supplier.
[ ]	**Abrasive paper**	Several	320- or 400-grit.	Lightly key steel bonding surfaces. Do not remove the magnet nickel coating.	Hardware supplier.
[ ]	**Lint-free wipes**	1 pack	Solvent-compatible.	Prevent recontamination after cleaning.	Electronics/industrial supplier.
[ ]	**Nitrile gloves**	1 box	Correct operator size.	Protect hands and keep skin oils off bond surfaces.	Pharmacy or safety supplier.
[ ]	**Mixing cups/sticks**	20+	Disposable and clean.	Enable accurate, contamination-free epoxy batches.	Hardware/craft supplier.
[ ]	**Dispensing syringes**	5-10	Disposable, blunt tipped.	Apply a repeatable thin adhesive layer.	Pharmacy/electronics supplier.
[ ]	**Polarity reference**	1	Known magnet with permanently marked N and S faces.	Prevents polarity errors during the alternating-pole build.	Prepare from one verified spare magnet.
[ ]	**Magnetic compass**	1	Small handheld compass.	Independent polarity verification before adhesive application.	Outdoor/electronics supplier.
[ ]	**Permanent markers**	2 colours	Solvent resistant where possible.	Mark pole identity and numbered magnet locations.	Stationery supplier.
[ ]	**Nonmagnetic tweezers**	1	Plastic, brass or suitable nonmagnetic tool.	Improve control and reduce finger trapping during placement.	Electronics supplier.
[ ]	**Magnet separators**	Many	Plastic, card or wood.	Keep loose magnets apart to reduce snapping and chipped plating.	Fabricate from scrap nonmagnetic material.
[ ]	**Masking tape**	1 roll	Medium tack.	Protects surfaces and supports temporary alignment fixtures.	Hardware/stationery supplier.

8. Electrical connection materials

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Flexible phase cable**	6 + spare	Initial choice: 16-18 AWG high-strand-count silicone cable; final gauge after measured current and temperature rise.	Carries three phase connections from each stator.	Communica, Mantech, RS or local electronics supplier.
[ ]	**Phase connectors**	6 pairs	Locking and touch-safe; rated above measured continuous current and DC bus voltage.	Permit independent stator connection and reconfiguration.	Electronics supplier.
[ ]	**Ferrules/terminals**	20 assorted	Match wire gauge and PCB terminal method.	Provide repeatable low-resistance terminations.	Electronics/electrical supplier.
[ ]	**Heat-shrink tubing**	1 kit	Assorted diameters and phase-identification colours.	Insulates joints and identifies phases/sensors.	Electronics supplier.
[ ]	**Strain reliefs**	4-6	Cable glands or clamps matched to cable OD.	Prevent cable motion loading PCB solder joints.	Electrical supplier.
[ ]	**Cable sleeve**	2-3 m	Expandable braid or spiral wrap.	Protects wiring and keeps it away from rotating parts.	Electronics supplier.
[ ]	**Cable ties/mounts**	1 pack each	Small, suitable for laboratory conditions.	Routes phase and sensor wiring securely.	Hardware/electronics supplier.
[ ]	**Temperature sensors**	6 rec.	Type-K thermocouples or suitable NTCs; at least three points per stator stack.	Monitor thermal distribution and protect the PCB windings.	RS, Communica or laboratory stock.
[ ]	**Sensor wire**	As req.	Twisted and insulated; compatible with chosen sensors.	Connects embedded temperature points to DAQ/logger.	Electronics supplier.
[ ]	**Protective-earth wire**	1-2 m	Green/yellow, about 2.5 mm2.	Bonds conductive frame and guard supports to protective earth.	Electrical supplier.

9. Drive and electrical protection

The inverter is intentionally not locked yet. First measure stator resistance and inductance and estimate/measure back-EMF; then size the controller and wiring.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Current-limited DC supply**	1	Adjustable 0-15 V initial commissioning supply; approximately 10 A or more; reliable current limiting essential.	Enables controlled low-energy electrical testing before full inverter operation.	Borrow from Power Lab if available.
[ ]	**Three-phase inverter**	1-2 ch.	Voltage/current ratings selected only after phase resistance, inductance and back-EMF are measured.	Commutates the motor and controls speed/torque.	VESC Project or laboratory inverter; do not buy a low-power board blindly.
[ ]	**Main DC fuse**	1 + spare	Above normal operating current but below cable/controller safe limit.	Protects the supply wiring and inverter from severe faults.	Automotive/electrical supplier.
[ ]	**Branch fuses**	2 + spare	One appropriately rated fuse per stator/controller branch.	Isolates a fault in one stage.	Electrical supplier.
[ ]	**Emergency stop**	1	Latching mushroom switch controlling a DC-rated contactor.	Allows immediate removal of motor power.	Automation/electrical supplier.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Main contactor/relay**	1	DC rated for the selected bus voltage and current.	Performs the actual power interruption commanded by the E-stop.	Automation supplier.
[ ]	**Pre-charge circuit**	1	Resistor/circuit matched to inverter DC-link capacitance.	Limits capacitor inrush and connector arcing.	Design after inverter selection.
[ ]	**Rotor-position sensor**	1 set	Encoder or three Hall sensors with adjustable bracket.	Provides commutation position and controlled start-up.	Select with controller.
[ ]	**Encoder target/magnet**	1	Compatible with selected encoder.	Creates the shaft-position reference.	Select with sensor.

10. Measurement and test equipment

Most items in this section should be borrowed rather than purchased for a single prototype.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Digital multimeters**	2	Suitable CAT rating and current/voltage range.	Continuity, phase resistance checks, voltage and sensor diagnostics.	Borrow from Power Lab.
[ ]	**Micro-ohmmeter / 4-wire meter**	1	Kelvin measurement capability.	Accurately measures low PCB phase resistance.	Borrow from Power Lab.
[ ]	**LCR meter**	1	Frequency range suitable for winding measurement.	Measures phase inductance.	Borrow from Power Lab.
[ ]	**Oscilloscope**	1	Sufficient bandwidth for inverter switching.	Observes phase voltage, back-EMF and switching waveforms.	Borrow from Power Lab.
[ ]	**Differential probe**	1	Voltage category/range above DC bus and switching transients.	Measures floating phase-to-phase voltages safely.	Borrow from Power Lab.
[ ]	**Current probe/sensors**	2-3	Range selected from predicted phase current.	Measures DC-link and phase current.	Borrow or instrument the prototype.
[ ]	**Temperature logger/DAQ**	1	At least six compatible channels preferred.	Records stator temperature simultaneously.	Borrow from Power Lab.
[ ]	**Optical tachometer**	1	Noncontact RPM measurement.	Independent verification of shaft speed.	Borrow from Power Lab.
[ ]	**Torque sensor/dynamometer**	1	Range selected from simulated torque with safety margin.	Measures output torque and supports efficiency calculation.	Power Lab/test facility.
[ ]	**Load cell and torque arm**	1 each	Known arm radius; load cell range selected from expected force.	Lower-cost reaction-torque method using T = F x r.	Power Lab or local supplier.
[ ]	**Gaussmeter**	1 if avail.	Probe suitable for surface-field comparison.	Checks magnetic pattern and compares individual magnets.	Borrow if available.
[ ]	**Accelerometer**	1 rec.	DAQ-compatible vibration sensor.	Detects imbalance, resonance and bearing problems.	Borrow from laboratory.

11. Assembly and alignment tools

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Digital caliper**	1	0-150 mm range; 0.01 mm display.	General dimensional inspection and stack measurement.	Workshop or tool supplier.
[ ]	**Outside micrometer**	1	Suitable range for 2 mm steel and spacers.	More accurate thickness measurement.	Workshop.
[ ]	**Dial indicator/base**	1 set	0.01 mm resolution or better; rigid base.	Measures shaft and rotor radial/axial runout.	Workshop.
[ ]	**Engineer's square**	1	Workshop-grade square.	Checks bearing/stator support perpendicularity.	Workshop.
[ ]	**Flat reference plate**	1	Surface plate or verified flat plate.	Provides assembly and alignment datum.	Workshop.
[ ]	**Torque screwdriver**	1	Range appropriate for M3/M4 fasteners.	Applies repeatable clamp torque without PCB damage.	Workshop/tool supplier.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Hex/screwdriver set**	1	Matches all fastener heads.	Assembly and service.	Workshop.
[ ]	**Soldering station**	1	Temperature controlled.	Phase-lead and temperature-sensor installation.	Electronics laboratory.
[ ]	**Stripper/crimper**	1 each	Matched to selected ferrules and wire gauge.	Reliable harness construction.	Electronics laboratory.
[ ]	**Arbor/bearing press**	1	Small controlled press with correct drifts.	Installs bearings without hammer impact or race damage.	Workshop.
[ ]	**Soft-jaw vice**	1	Nonmarring jaws.	Holds hubs and brackets safely.	Workshop.
[ ]	**Deburring tools**	1 set	Files, countersink and edge deburrer.	Removes sharp laser-cut/machined edges.	Workshop.
[ ]	**Static balancing jig**	1	Low-friction knife-edge or bearing-based setup.	Checks each magnetised rotor before powered testing.	Fabricate or use workshop equipment.
[ ]	**Small digital scale**	1	Fine resolution appropriate for magnet/adhesive mass.	Helps control adhesive mass and investigate imbalance.	Laboratory/tool supplier.

12. Safety enclosure and PPE

The 3 mm guard thickness is a starting specification, not a certified containment rating. Maximum permitted speed should remain conservative until rotor stress, retention and balance are verified.

Tick	Item	Qty	Specification / dimensions	Function and role	Source / action
[ ]	**Polycarbonate sheet**	Sized to rig	Clear polycarbonate; 3 mm minimum starting point for prototype guard. Final thickness/spacing must follow maximum-speed risk assessment.	Prevents contact and helps contain fragments. Use polycarbonate rather than brittle acrylic/Perspex.	Maizey polycarbonate
[ ]	**Guard frame**	1	Aluminium angle or 2020 extrusion, mechanically rigid.	Supports transparent panels around all rotor planes.	Maizey/hardware/automation supplier.
[ ]	**Guard fasteners**	As req.	M4/M5 bolts with large washers; avoid point loading sheet edges.	Secure panels without initiating cracks.	Fastener supplier.
[ ]	**Guard interlock**	1 rec.	Microswitch or safety switch wired into contactor control.	Prevents energisation while the guard is open.	Automation supplier.
[ ]	**Warning labels**	1 set	Rotating machinery, strong magnets and electrical hazard.	Warn operators and visitors.	Safety supplier or print locally.
[ ]	**Eye/face protection**	Per person	Impact-rated safety glasses; face shield during early commissioning if risk assessment requires it.	Reduces injury risk during rotor assembly and initial spin tests.	UCT PPE stores or safety supplier.

13. Workshop fabrication request

The workshop package should ultimately contain dimensioned drawings for the following parts:

- Two 2 mm mild-steel rotor back-iron discs.
- Three magnet-retention plates, one for each active magnet face.
- Two 8 mm-bore split-clamp rotor hubs.
- Four stator clamp plates.
- Twelve stator compression sleeves.
- Two bearing housings/supports for 608 bearings.
- Two adjustable stator-support frames.
- One rigid baseplate and bearing brackets.
- Required rotor/stator axial spacers.
- One encoder mount and one torque-arm/load bracket.
- One polycarbonate safety enclosure and support frame.

14. Information to freeze before final manufacture

[ ] Optimised magnet centre radius.

[ ] Confirmed rotor outside diameter and adequate edge margin.

[ ] Hub pilot, flange and bolt-circle geometry.

[ ] Complete axial stack length, including all air gaps and spacers.

[ ] Bearing-support spacing and final shaft length.

[ ] Air-gap adjustment range and mechanical hard stops.

[ ] Encoder, coupling and guard clearances.

[ ] Maximum intended operating speed and rotor-retention safety factor.

15. Recommended procurement sequence

Priority	Action	Items
A - Obtain now	Buy/borrow items unaffected by final CAD	8 mm shaft stock; 608-2RS bearings; 8 mm clamp collars; prototype M3/M4 hardware; nylon washers; feeler gauges; shims; slow-cure epoxy; cleaning supplies; wiring; temperature sensors; guard material.
B - Optimise	Complete electromagnetic and mechanical decisions	Magnet radius; rotor OD; air-gap range; axial-force result; maximum speed; controller requirements.
C - Freeze CAD	Issue one complete assembly and part drawings	Hubs; rotors; retainers; stator clamps; bearing blocks; supports; base; spacers; shaft length; guard.
D - Fabricate	Manufacture and inspect parts	Check dimensions, flatness, concentricity, deburring and material certificates/identification where relevant.
E - Assemble safely	Build in controlled stages	Dry fit; stator stacks; bearing/shaft alignment; magnet polarity jig; adhesive coupons; rotor bonding; retention plates; static balance; low-speed guarded test.

16. Final assembly gate

Do not begin powered testing until:

The rotors are mechanically retained and statically balanced; shaft runout and air gaps have been measured around a full revolution; all fasteners are secured; phase circuits pass continuity/insulation checks; sensors and E-stop operate correctly; and the complete rotating assembly is enclosed by the guard.

Document note: supplier links are indicative starting points. Confirm current stock, exact suffixes, dimensions, ratings and compatibility before ordering. Provisional geometry must be replaced with released CAD dimensions.